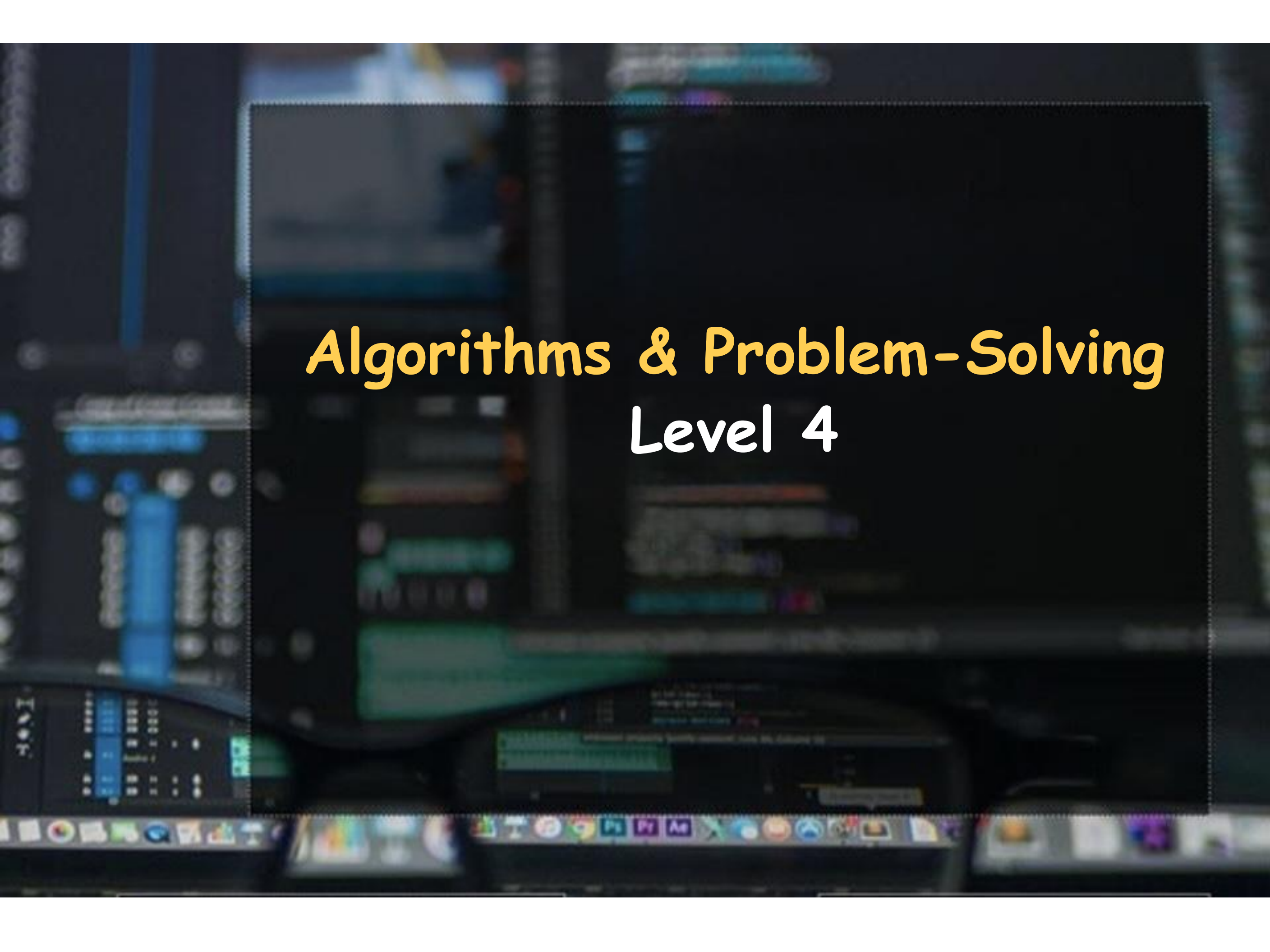
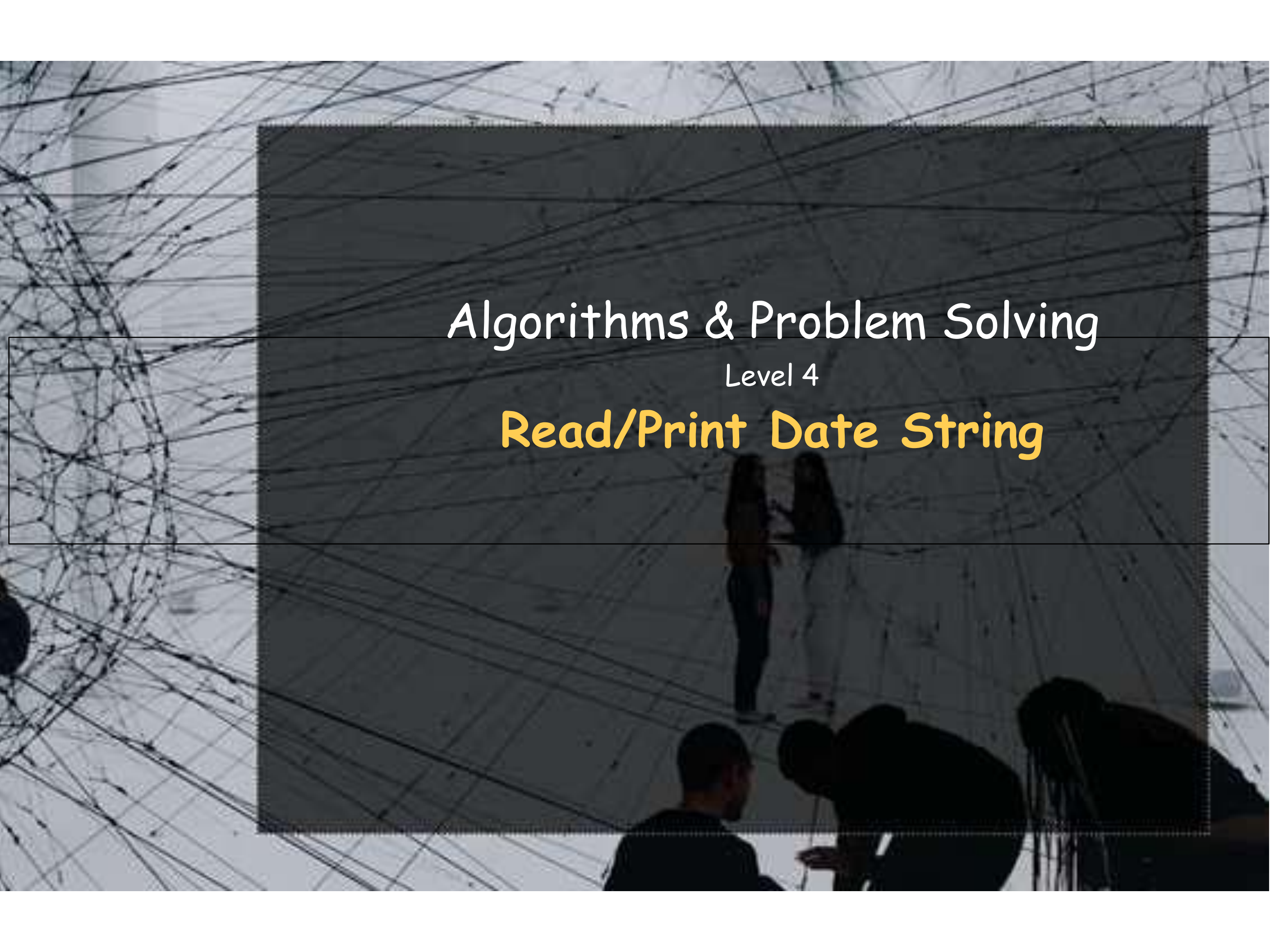




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, adding a human element to the technical theme.

Algorithms & Problem Solving

Level 4

Read/Print Date String



Definition

- A program to read a date in string format, such as 31/3/2022, convert it to a data structure (Struct) containing the day, month, and year, then print it separately, and then convert the structure back to text again.



Explanation:

- The program's objective is to handle dates in a dual manner (as text and as a data structure).
- The program uses several functions to split the text, convert between text and structure, and print.

Procedures

1 Define a struct named sDate

- It contains three values:

 - short Day

 - short Month

 - short Year

2 SplitString Function

- Definition: Splits a string into parts based on a delimiter

- Used here to separate the day, month, and year when a / is present

- 📌 Note: The SplitString function returns a vector (vector<string>) containing the separated parts

3 StringToDate Function

- Definition: Converts a string to a variable of type sDate

- Relies on SplitString to extract the parts

- Uses stoi to convert the strings to numbers

- 💡 Example: "31/3/2022" → Day=31, Month=3, Year=2022

4 DateToString Function

- Definition: Returns the date structure to string separated by a marker /

- 💡 Example: (31, 3, 2022) → "31/3/2022"

5 ReadStringDate Function

- Definition: To read text from the user

- Uses getline(cin >> ws, dateString) to avoid problems caused by spaces

⑥ Main Function (main())

- Reads the date text from the user
- Converts the text to a date using `StringToDate`
- Prints the day, month, and year separately
- Then prints the reconverted text using `DateToString`



Example:

Please Enter Date dd/mm/yyyy? 31/3/2022

Day: 31

Month: 3

Year: 2022

You Entered: 31/3/2022.

Summary

- ✓ The program explains how to handle dates between strings and structures.
- ✓ It relies on string splitting (SplitString) and transformation (stoi and to_string). ✓
- ✓ This facilitates later date comparison or performing time operations on them.

Thank you 😊